PARISHRAM 2.0

CLASS 12TH 2026

Electric Charges and Fields

PHYSICS LECTURE-1

BY-DEV AGNANI SIR



TOPICS TO BE COVERED

1. Electric Charges
2. Properties of Electric Charge
3. Coulomb's Law } Introduction

केन्द्रीय माध्यमिक शिक्षा बोर्ड
दिल्ली
असतो मा सद्गमय

Fielding set hai tumhari...

Tension lene ka nahi re baba, Tension dene ka..



CLASS-XII
Board exams

Ending kharaab hai na?



Ending abhi aayi hi kaha hai?

Apni kahani Hai!

Ending change kar lenge!!!

Class-11th Barbaad !!! 🤡
Class-12th mein esa matt hone dena !!! 🔥

↳ PADHAI POINTERS :—

- Notes Class ke sath Banane Hai!
  - ↳ Class ke baad Complete notes.
- DADA method : DAmdar Dev Agnani Method.
- under 24 Hrs → Notes Revision.

→ ZKTK Rule. | → समय के साथ चलो!



Class-XII CBSE board Exam Chapter-wise blueprint

Unit	Chapters	Marks (70)
(I) Electrostatics	**Ch-1:** Electric Charges and Fiels 5	16
	Ch-2: Electrostatic Potential and Capacitance 5	
(II) Current Electricity	**Ch-3:** Current Electricity 6	
(III) Magnetic Effects of Current and Magnetism	**Ch-4:** Moving Charges and Magnetism	17
	Ch-5: Magnetism and Matter	
(IV) Electromagnetic Induction and Alternating Currents	**Ch-6:** Electromagnetic Induction (EMI)	
	Ch-7: Alternating Current (AC)	
(V) Electromagnetic Waves	**Ch-8:** Electromagnetic Waves (EM Waves)	18
(VI) Optics	**Ch-9:** Ray Optics and Optical Instruments	
	Ch-10: Wave Optics	
(VII) Dual Nature of Radiation and Matter	**Ch-11:** Dual Nature of Radiation and Matter	12
(VIII) Atoms and Nuclei	**Ch-12:** Atoms	
	Ch-13: Nuclei	
(IX) Electronic Devices	**Ch-14:** Semiconductor Electronics	7

Electric Charges and Fields

CBSE website Syllabus Class-XII. → introduction.

➢ Electrostatics

- Electric charges, Conservation of charge, Coulomb's law-force between two- point charges, forces between multiple charges; superposition principle and continuous charge distribution.
- Electric field, electric field due to a point charge, electric field lines, electric dipole, electric field due to a dipole, torque on a dipole in uniform electric field.
- Electric flux, statement of Gauss's theorem and its applications to find field due to infinitely long straight wire, uniformly charged infinite plane sheet and uniformly charged thin spherical shell (field inside and outside).



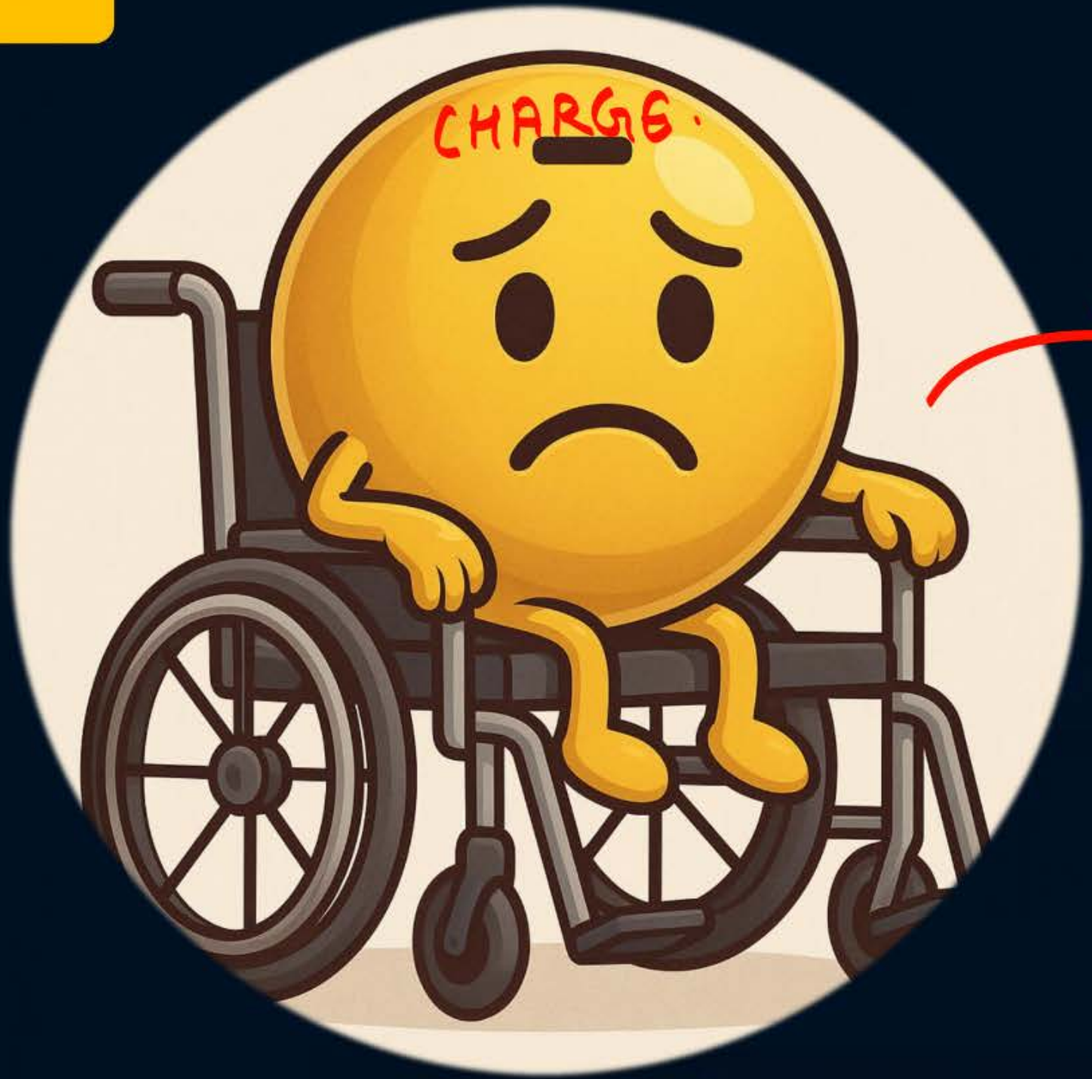




Electrostatics

UNIT

Study of charge at rest.



Can't pursue motion

STATIC

Electric Charges



Definition of Electric charge

- Electric Charge is an intrinsic property of all the elementary particles such as protons, electrons, etc. of which all the matter around us is made up of.
- It is because of this charge only that the particle is able to produce as well as experience the electric and magnetic effects.

Characteristics of Electric charge

→ It is a Scalar Quantity.

→ Charge is always associated with mass.

Some important Values

- Charge on an electron (e) = -1.6×10^{-19} C
- Charge on a proton (p) = 1.6×10^{-19} C
- Mass of an electron = 9.1×10^{-31} Kg
- Mass of a proton = 1.6×10^{-27} Kg
- Mass of a neutron = 1.6×10^{-27} Kg

Same

Types of Charges :-

1. Positive Charge. ⟶ It has more p^+ than e^-.
2. Negative Charge. ⟶ It has more e^- than p^+.

→ Like charges repel from each other.

→ Unlike charges attract each other.

e^- ⇒ Negatively charged.

p^+ ⇒ Positively charged.

p^+ p^+ } → repel

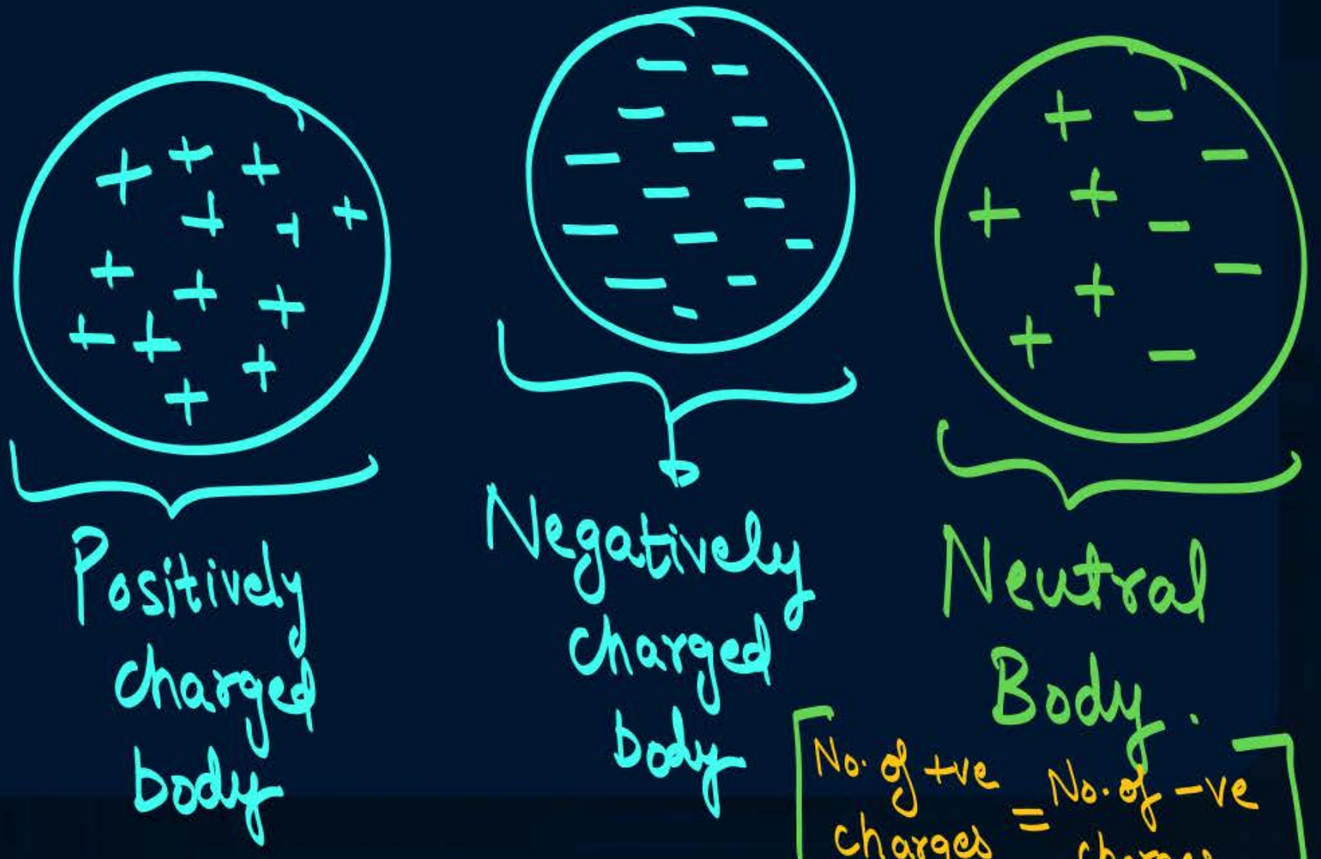
e^- e^- } → repel

p^+ e^- } → Attract.

How can we charge a body?

- There are **three main methods of charging** a body in physics (especially electrostatics). These methods explain how an object can gain or lose **electric charge**:

- Charging by Friction
- Charging by Induction
- Charging by Conduction



Conductors and Insulators

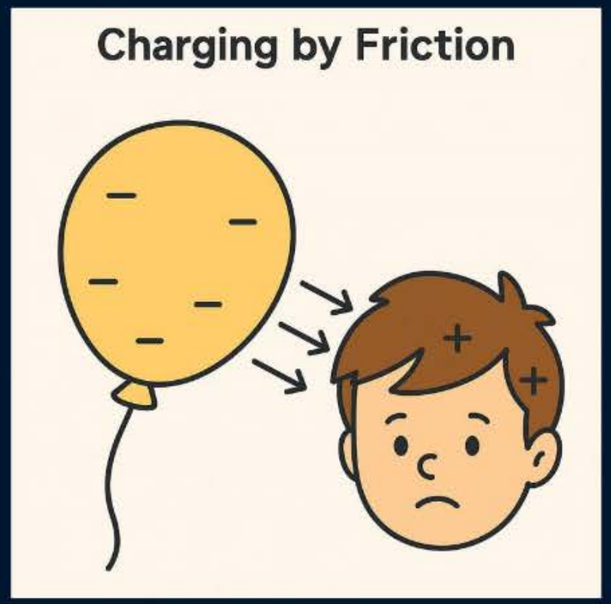
Class - 9th

Conductor	Insulator
Allows electric current to flow	Does **not** allow current to flow
Has **free electrons**	Has **no free electrons**
Low resistance	**High resistance**
Example: Copper, Aluminum	Example: Rubber, Plastic
Used to **carry** electricity	Used to **block** electricity

Some particles exhibit both Conductor and Insulator properties, called as **Semi-Conductors**

Charging by Friction (Rubbing)

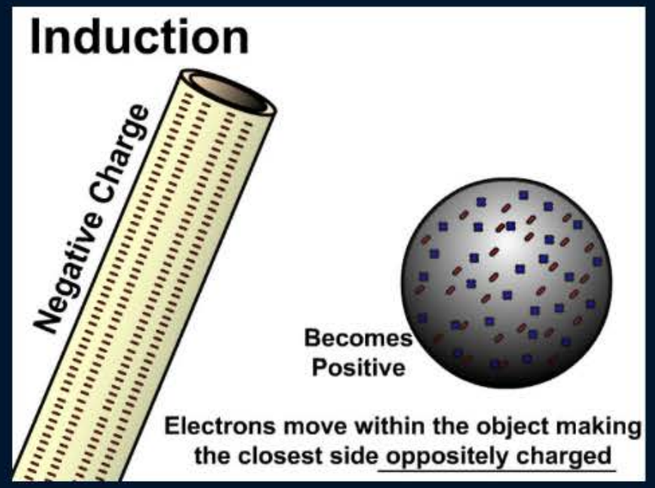
- When two different objects are rubbed together, electrons are transferred from one to the other.
- One becomes positively charged, and the other becomes negatively charged.
- Type of charge transfer: Electron loss or gain.



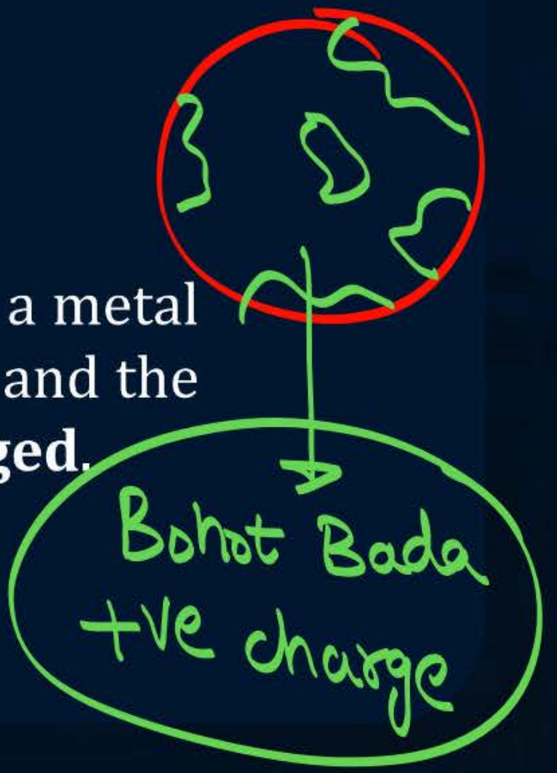
E.g. Rubbing a balloon on your hair makes it negatively charged.

Charging by Induction (without contact)

- A charged object is brought near a neutral conductor (without touching).
- The presence of the charged object rearranges electrons in the neutral object.
- Earthing or grounding is used to allow or remove electrons, leaving the object charged opposite to the nearby charge.
- Type of charge transfer: Electron redistribution followed by grounding



E.g. A negatively charged rod is brought near a metal sphere → electrons move away from the rod, and the sphere is earthed to become **positively charged**.



Charging by Conduction (contact)

- What it is: A charged object is touched to a neutral object.
- Charge flows from the charged object to the neutral one.
- The neutral object ends up with the same type of charge as the charged one.
- Type of charge transfer: Electrons move through contact



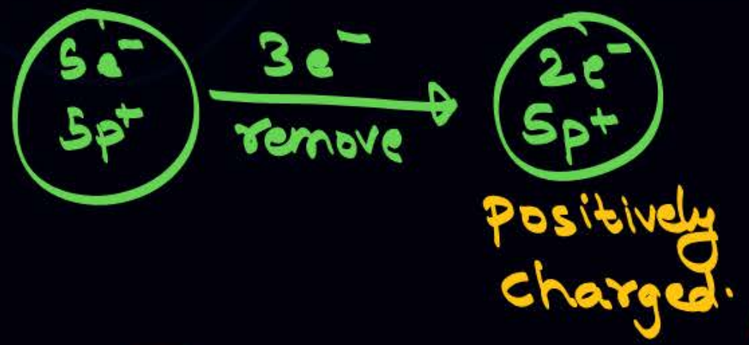
E.g. Walking on Carpet, Then Touching a Doorknob:

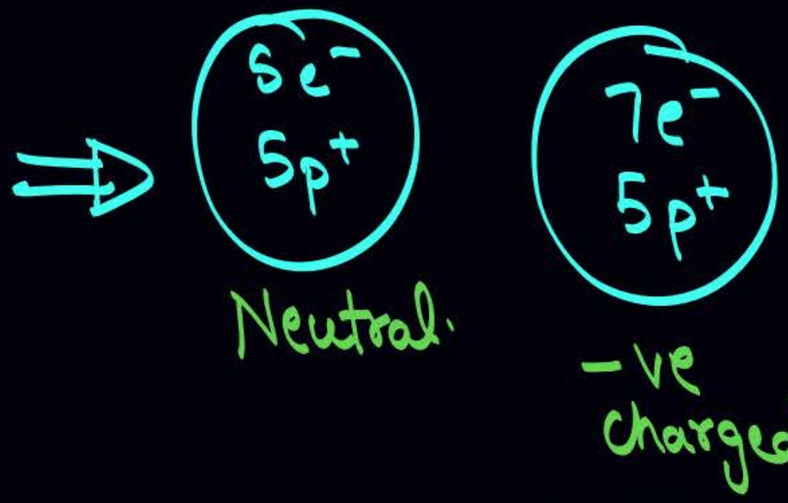
- You accumulate negative charge through friction (your body gets charged).
- When you touch a metal doorknob (conductor), some of that charge transfers to it — sometimes you feel a shock!

Mass

e^- and p^+ have mass.







DADA

Protons are never transferred, or we can say that protons are never added or removed from a body.

QUESTION

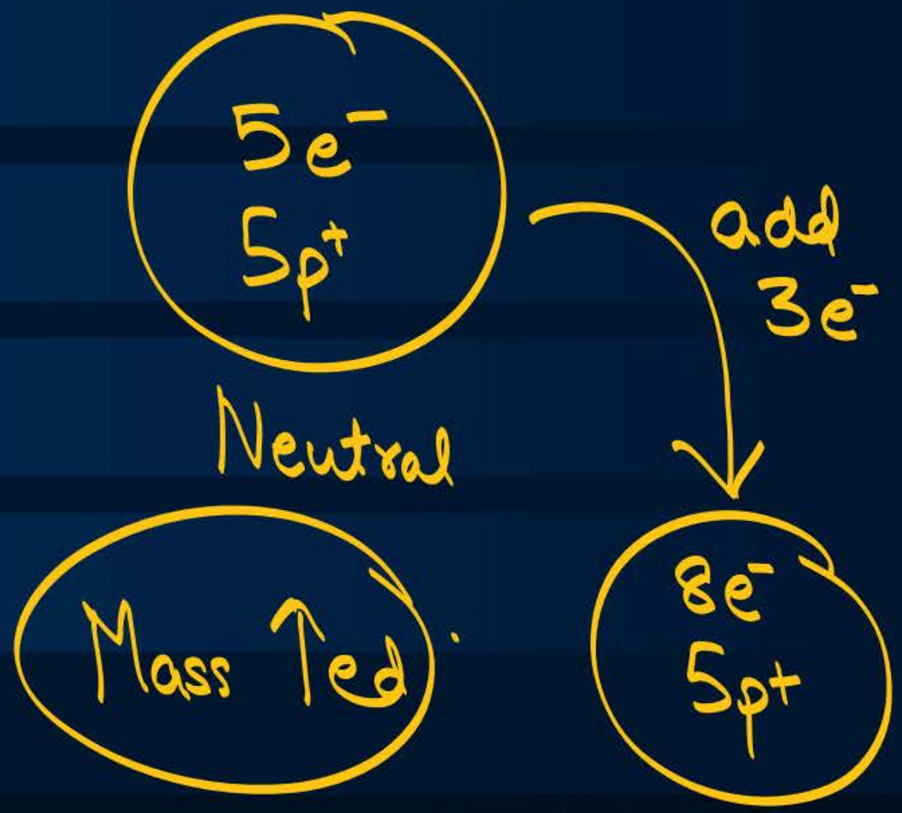
A neutral body is charged negatively, what happens to its mass?

A Mass is increased

B Mass is decreased

C Remains same

D Can't say



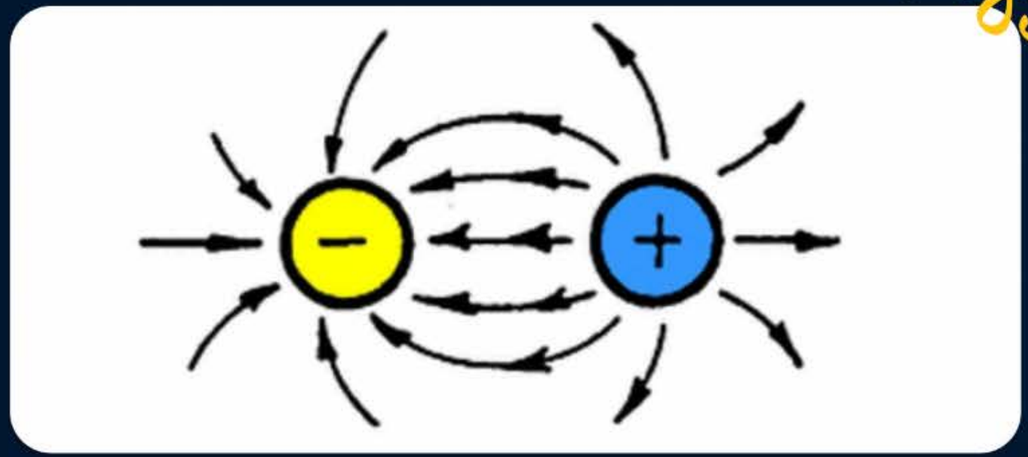


Properties of Electric Charge

- Additivity of Charges
- Conservation of Charge
- Quantization of Charge

→ Charge is invariant

→ Accelerated charge radiates energy.



QUESTION

Which of the following is not a property of electric charge?



A It is Scalar

B It is Vector

C It is Conserved

D It is quantized

THANK YOU

